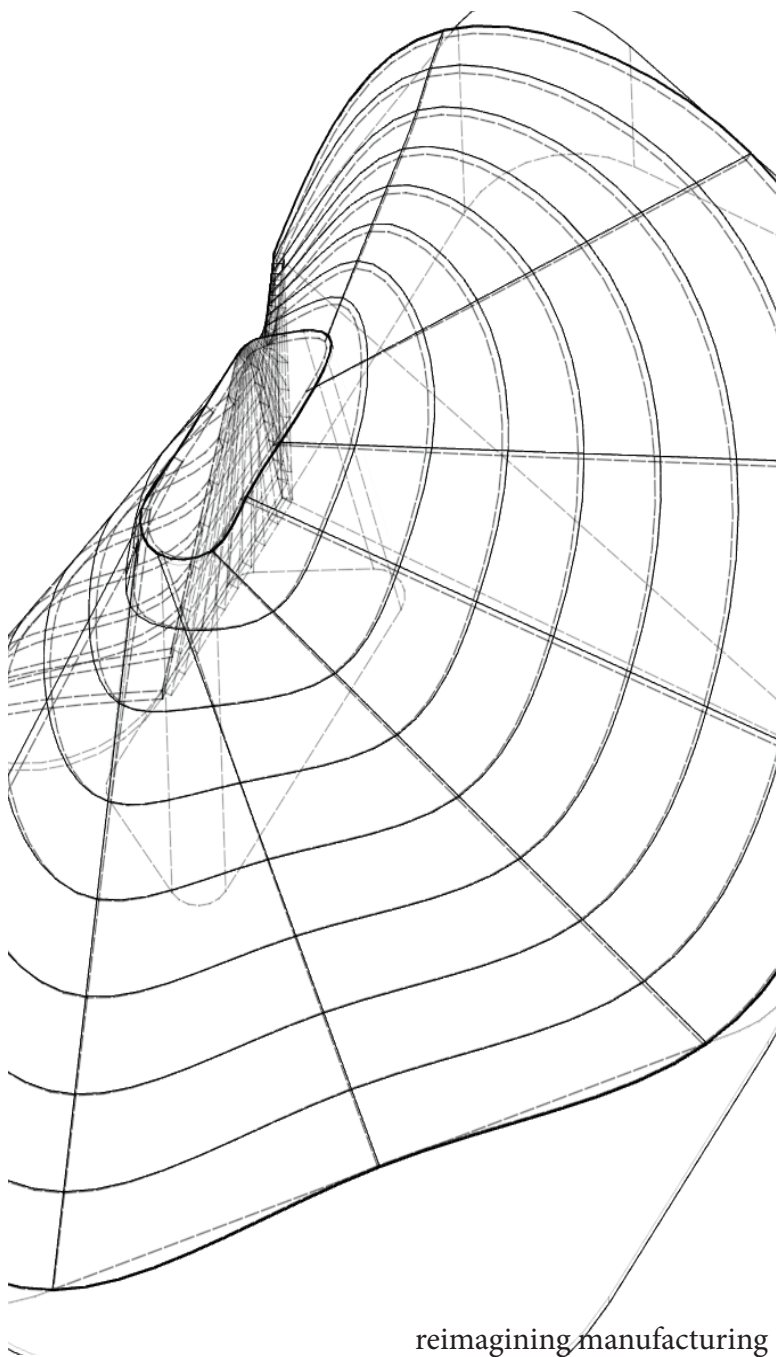




reimagining manufacturing

How We May Make

This thesis proposes a transformative approach to hardware production, envisioning a future where devices are made on-demand at local nano-factories, tailored to individual specifications. Traditionally, sustainability has been relegated to an afterthought in manufacturing. However, this research leverages emerging technologies to fundamentally integrate sustainable practices into the core of design and production processes, using advanced materials and methods. To illustrate the practical applications of this model, a computer mouse was designed and produced, serving as a ‘boundary object’ to spark crucial conversations among researchers, designers, technology industry, government agencies, and the public.

ABSTRACT

Consumer hardware design and manufacturing are at a critical juncture. The hardware industry's reliance on centralized production, long supply chains, and linear life cycles has made it highly vulnerable to disruptions due to climate change [1] [2]. These issues call for a radical rethink of how new devices are designed, produced, and recycled. This thesis addresses this urgent challenge by investigating the potential of volumetric additive manufacturing (VAM) to create personalized, sustainable, and locally manufactured hardware products.

At its core, VAM represents a shift from traditional layer-by-layer additive manufacturing to a faster, more efficient, and waste-free production process capable of producing intricate forms in minutes [3]. VAM's potential to create intricate multi-material forms without traditional assembly processes offers a promising direction for a more sustainable hardware ecosystem.

To explore the possibility of creation of hardware, this research undertook the prototyping of a simple conductive pathways overmolded on passive electronic components using VAM. Further, to serve as a fully functional vision demonstration for the technology proposed, a 3D-printed computer mouse was designed and prototyped.

VAM enables localized manufacturing, reducing reliance on fragile supply chains and minimizing carbon emissions associated with shipping and raw material waste. Furthermore, it offers a path to improved recyclability and resource recovery, addressing key limitations of current design practices. However, limitations remain, particularly in integrating silicon-based semiconductors and scaling VAM to multi-material complexities demanded by existing hardware devices. The work identifies potential in materials science, generative design tools, and advanced recycling technologies, calling for further research and collaboration across disciplines.

By envisioning a future where hardware devices are produced on-demand, tailored to individual users, and optimized for circular economies, this thesis serves as both a speculative exploration and a call to action. It urges designers, engineers, researchers, and policymakers to reimagine the hardware industry for a climate-conscious and resilient future.

How We May Make

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INTRODUCTION

The consumer hardware industry has long operated on a model of global supply chains and mass production, optimizing for cost efficiency at the expense of the environment. While this system has successfully delivered technological innovations to billions of consumers, it has also emerged as a significant contributor to climate change. Consumer hardware manufacturing emissions are expected to reach 852 million metric tons of CO₂ annually by 2030 without intervention [4]. Despite the mounting evidence of environmental impact, the industry's response has largely been limited to superficial improvements and incremental changes, such as slightly more efficient manufacturing processes or marginally increased use of recycled materials.

The disconnect between the industry's actions and the urgency of climate change is profound. Current sustainability initiatives primarily focus on end-of-life solutions like recycling and repairability. While beneficial, these measures fail to address the fundamental problems embedded in the industry's DNA: resource extraction, carbon-intensive manufacturing, and global transportation networks. This disconnect is particularly evident in the electronics sector, where complex supply chains involve over on average 3,000 discrete processes and rely heavily on environmentally destructive practices.

The stakes of this challenge have never been higher. As climate change manifests through devastating weather events, the need for systemic change becomes increasingly urgent. This thesis proposes a radical reimagining of hardware production through volumetric additive manufacturing, offering a path to sustainable, localized manufacturing that could significantly reduce environmental impact while maintaining technological progress.

Through the development of Model A, a concept for a fully volumetrically printed computer mouse, this work offers a tangible vision of how hardware could be produced, distributed, and reclaimed in a sustainable future. This is speculative but technically grounded path forward challenges designers and manufacturers to fundamentally rethink not just how the entire hardware ecosystem could be restructured.

BACKGROUND

For much of the late 20th century, the manufacturing industry revolved around one primary goal: the efficient extraction of atoms and their recombination into products that could be shipped across the globe at the lowest possible cost. These systems undoubtedly added value to consumers' lives, but the environmental cost of this progress was largely ignored. Victor Papanek, a pioneer in sustainable design, sounded an early warning in his book *Design for the Real World* (1971), critiquing industrial design's environmental toll. However, it wasn't until the Intergovernmental Panel on Climate Change (IPCC) released its first report in 1990 that the narrative shifted [5]. What had once been viewed as local pollution crises was reframed as a global challenge. Climate change revealed the interconnectedness of human activity, where emissions saw no borders and consequences rippled across the planet.

Despite this newfound awareness, the manufacturing and technology sectors largely failed to recalibrate their practices. Environmentalism was not seen as a moral imperative but as a strategic inconvenience. It was only when reframed as an opportunity for efficiency—keeping resources in a closed loop—that sustainability gained mainstream traction. Kenneth Boulding's concept of the "closed economy" in his 1966 essay *The Economics of the Coming Spaceship Earth* underscored the importance of cycling materials within finite systems. Building on this, Walter Stahel introduced the notion of extending the life cycle of goods in the 1980s [6], a concept that the Ellen MacArthur Foundation later amplified into the framework of the circular economy [7]. This model champions recycling, reparability, and product longevity as pathways to sustainability. Policymakers have embraced these ideals, exemplified by the European Union's stringent regulations on standardising connectors and reparability for all devices. But this approach of trying to solve for climate change through regulation stifles innovation. Mandating universal standards like USB-C may reduce waste, but it constrains progress. What if a startup invents a USB-X connector that's exponentially faster and more energy-efficient? The regulation inhibits its adoption, locking industries into present-day standards.

Such initiatives are lauded as wins for sustainable hardware. Yet, these measures fall short in addressing the urgent reality of climate change. Recycling and reparability, while beneficial, are insufficient when juxtaposed with the scale of the problem. Retrofitting sustainability into an entrenched system of extraction, production, and disposal is like patching a leaking boat while it's already sinking. As Boulding might have argued, the problem isn't just with the materials but with the system itself. Consider for example modularity and reparability, hailed as cornerstones of sustainable hardware. These principles aim to combat planned obsolescence—the intentional shortening of product life cycles—but they overlook "unplanned obsolescence." As user needs and technological capabilities advance exponentially, even the most durable hardware can become functionally irrelevant. A repairable, modular phone may outlast its non-repairable equivalent, but it will still be discarded when its outdated components can no longer support evolving user demands. Mandating reparability also increases production costs, which are passed on to consumers, which could limit accessibility of these technologies to the Global South.

A system optimized for extraction, production and global dissemination resists the transformative change that sustainability demands. This shift requires abandoning the retrofitting mindset and instead asking radical questions about the very nature of production and consumption. Only by stepping back from the established model can we design for a future where technology serves humanity without imperiling the planet.

MOTIVATION

In August 2018, unprecedented monsoon rains resulted in severe flooding across Kerala, including my home, where my family had resided for over two decades. This event marked the first time we were forced to evacuate and seek refuge with relatives. The floods caused extensive damage to my home and the community that year, and in the years since, extreme precipitation events have become almost commonplace in my state, with devastating floods occurring roughly every other year. Climate change was no longer an abstract concept for me—it had altered lives, homes, and neighborhoods.

The climate crisis is here, now, and demands immediate, systemic action. The consumer hardware industry, which forms the backbone of modern technology, is a significant contributor to the problem. Manufacturing of new electronic products, especially information and communications technology (ICT) devices alone accounts for 580 metric ton of greenhouse gas emissions in 2020. Yet, despite its outsized role, the sector remains alarmingly complacent, caught in a cycle of superficial greenwashing and incremental, self-congratulatory changes.

As a hardware design engineer, I have seen this complacency firsthand. Annual environmental reports from major corporations boast of switching to slightly recycled materials or marginally improving manufacturing efficiency. These changes, often marketed as groundbreaking, fail to address the deep systemic flaws embedded in the industry's operations. The supply chains that underpin our devices rely on unethical practices: exploitative labor conditions, environmentally destructive mining, and fossil-fuel-intensive logistics networks. Reports like the Amnesty International study on cobalt mining in the Democratic Republic of Congo [8] reveal the human toll of the industry's insatiable appetite for raw materials.

The production process itself exacerbates the issue. It takes over 3,000 discrete steps to transform raw materials into the sleek gadgets we love. Each step, from extraction to assembly, contributes to environmental degradation and carbon emissions. The logistics of moving components around the globe rely on cheap, carbon-intensive transportation networks. Meanwhile, low wages in developing countries perpetuate cycles of poverty, ensuring that the true costs of production are borne by the most vulnerable.

Even as an environmentally conscious hardware designer, I am often left with no choice but to rely on this entrenched system. The myopic development cycles of the technology industry leave little room for innovation outside its preordained framework. The designer's toolkit is not just comprised of computer-aided engineering (CAE) tools or the rapid prototyping and production tools but the fossil-fuel-powered logistics, exploitative labor practices, and unsustainable mining and production methods. This toolkit enable designers to churn out products with minor, iterative improvements that perpetuate the cycle of consumption and waste.

This is why I am compelled to act—not only as a professional in the field but also as someone whose life has been directly shaped by the consequences of inaction. The hardware industry, for all its ingenuity, remains shackled to an outdated system optimized for profits. To continue leveraging this system is to perpetuate harm—to the planet, to the people who make our devices, and to those who suffer from the environmental fallout of our choices.

I seek to move beyond retrofitting sustainability into existing systems. What if we could explore advanced manufacturing methods such as volumetric 3D printing to create devices that align with a future where environmental impact is not an afterthought but a core design principle.

This project is a call to action. It challenges the hardware industry to acknowledge its role in the climate crisis and embrace the opportunity invest in transformative change.

APPROACH

At its core, manufacturing is about manipulating atoms into precise arrangements to create functional products. The goal has always been to do this as efficiently and cost-effectively as possible. Yet, when sustainability practices are retrofitted into this equation, they often feel like an afterthought, clashing with manufacturing's fundamental principles. What if, instead of forcing sustainability into the existing system, we reimagine the process itself—starting from first principles?

This approach is no longer theoretical. Advances in additive manufacturing (AM), or 3D printing, have revolutionized the fabrication of polymers and metals and are now extending to ceramics and composites. Multimaterial and multistep printing are maturing rapidly, enabling the fabrication of functional components within a single manufacturing process. For example, researchers have demonstrated the ability to additively produce printed circuit boards (PCBs), along with active and passive components such as capacitors, resistors, inductors[9] [10], and even OLED displays[11].

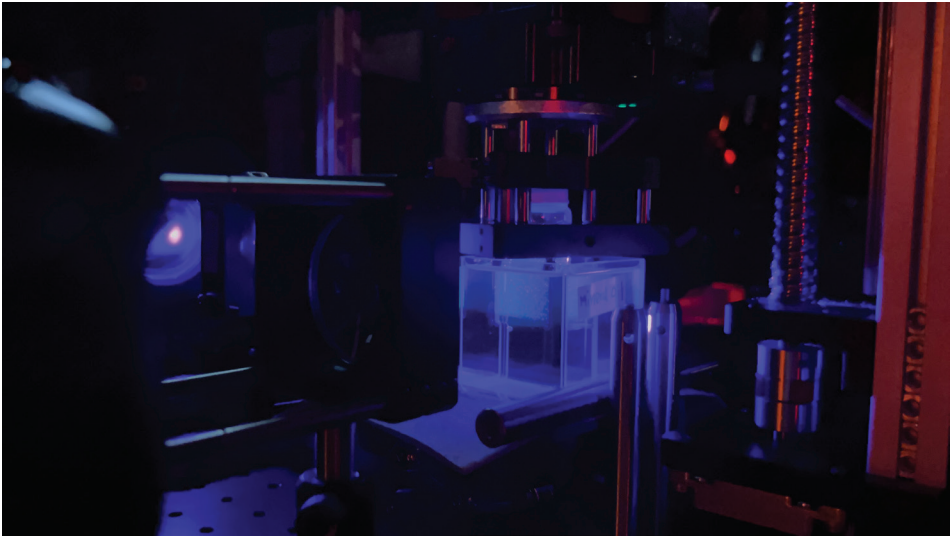
Additive manufacturing inherently minimizes material waste, a critical advantage over traditional methods such as injection molding and milling, which can result in material losses averaging 20–30%. By contrast, AM theoretically achieves near-zero material waste during production, drastically reducing the environmental footprint associated with lost material. Yet, the industry has been slow to adopt these technologies at scale, primarily due to limitations in production speed. Layer-by-layer 3D printing, the most common AM method, remains slow relative to conventional techniques. However, volumetric additive manufacturing (VAM) offers a breakthrough. Unlike traditional AM, VAM fabricates an object in one continuous process using volumetric light projections, achieving speeds that are up to 100 times faster for equivalent volumes. UC Berkeley's Nano Manufacturing Lab, led by Professor Hayden Taylor, has pioneered a version of VAM called Computed Axial Lithography (CAL)[12]. The process can produce polymer and ceramic components with a volume of up to 10 cubic centimeters in under 15 seconds at fine resolutions.

To explore the feasibility of using CAL for printed electronics, I worked with the Nano Manufacturing lab. My first experiment involved over-molding a 0.5W LED with dipentaerythritol pentaacrylate (DPPA) resin, leaving a cavity for conductive material. I used liquid gallium to create a conductive path, successfully lighting the LED with a coin cell battery. For comparison, conventional PCB fabrication would have required multiple steps—etching, drilling, and layering—and significantly more time, materials, and waste.

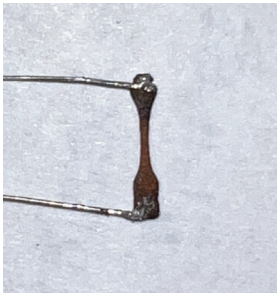
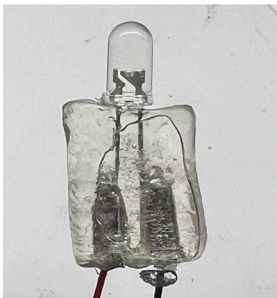
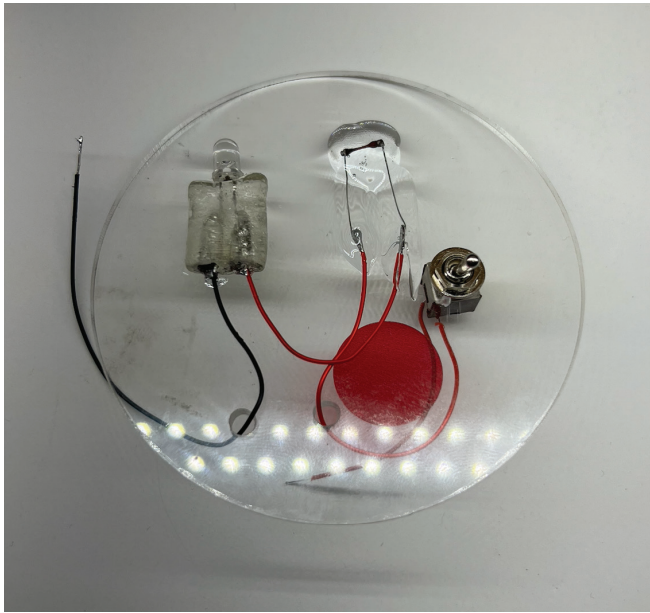
I also experimented with creating a functional circuit with a copper dog-bone printed using Hydrogel Infused Additive Manufacturing (HIAM), a method developed by Taylor Waddell et al. [13]. This process involves infusing a hydrogel with metal salts and reducing the salts to form conductive paths. Success of this underscored the feasibility of creating electronic circuits additively, without the need for traditional PCB manufacturing processes.

These experiments were more than proof-of-concept; they were an inquiry into whether fundamental physical or mechanical constraints might limit the vision of a fully volumetrically printed hardware device. Conversations with experts like Johanna Schwartz at Lawrence Livermore National Laboratory further clarified that no insurmountable barriers exist—only challenges requiring ingenuity and refinement.

To advance this research beyond theoretical discussions and into practical application, a 3D-printed computer mouse was specifically designed to conform to the ergonomics of my own hand. This prototype was made using conventional layer-by-layer resin printing processes on to which embedded electronic components were placed. It is intended to showcase the adaptability of 3D printing techniques and provided a personalized example of how such technology can be used to create custom, user-specific products.



CAL Printing test parts at the Nano Manufacturing Lab, Berkeley



First test print with in-situ Gallium Filled Conductive Path (top right) and HIAM Copper Conductor (bottom right)

OUTCOME

The computer mouse became the stage for my vision of sustainable, fully additive manufacturing. The prototype is called Model A, drawing inspiration from a hearing aid of the same name, the first mass-produced consumer hardware product with embedded electronics.

Prototyping Model A

As a designer who spends countless hours using CAD tools, I wanted this mouse to be deeply personal, and optimized for my specific ergonomic needs. The design process began with physical exploration: taking impressions of my palm and analyzing my natural grip and hand posture during use. These observations informed the prototype's shape and switch positioning, refined through iterative paper models. Each mockup was tested for comfort and usability, tweaking angles and positions until the form felt intuitive. Satisfied with the physical design, I translated it into a detailed CAD model. For the electronic components, I reverse-engineered the circuit from a Logitech B100 mouse, using a breadboard to validate its functionality. The CAD model was updated with cavities to house the salvaged components and pathways for conductive traces.

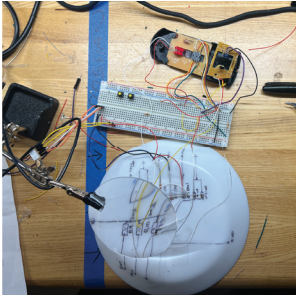
The initial physical prototype of the mouse was produced using a Form 3 SLS printer with clear resin. Once the printing was completed, the circuit components of the B100 model were meticulously integrated into their specific cavities within the mouse. For the conductive connections, I utilized tin-coated 26 AWG wires, which were carefully soldered onto the components mounted on a ceramic base. After assembling and several rounds of iterative debugging, the mouse was successfully tested for functionality on a MacBook Pro, confirming its operational readiness.

Through this process, I was also exploring the constraints of current technologies and tools, and how we could overcome them. The following were the questions posed to industry experts and researchers-

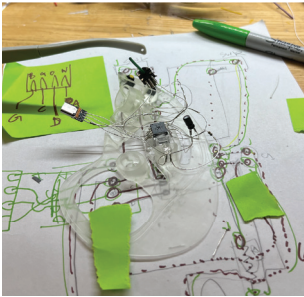
- Circuit Design Beyond Two Dimensions: Traditional PCBs are flat, but additive manufacturing unlocks the entire 3D space for circuit design. How might we rethink circuit layout tools to take advantage of this freedom? Could generative design algorithms optimize pathways for both performance and structural support?
- Structural Conductors: Could conductive pathways double as structural elements, merging electrical and mechanical functionality?
- Material and Process Constraints: What materials would make environmental sustainability or even bio-degradable?

Why this artifact matters?

Model A is a boundary object [14], created is to catalyze dialogue and experimentation. It highlights gaps in research, design tools and manufacturing techniques that need addressing to fully realise the vision. It invited researchers, designers, tech executives, and policymakers to reconsider what hardware design could look like if sustainability were to be baked into its design and production process.



Reverse-engineering the Logitech B100 mouse and transferring components to a ceramic dish.



Placing components in the cavities to be overmolded.



Model A, in two parts.

DISCUSSION

Every industry bears responsibility for mitigating its contributions to climate change, and this thesis is my attempt to redirect the hardware sector's attention toward a fundamental shift to sustainable production. Yet, the implications of this work go beyond the ethical imperative of addressing climate change. They also touch on the economics and logistics of hardware production, suggesting a path forward that is both sustainable and resilient.

A Call to Action for Resilience

For skeptics who may not see this as a climate action manifesto, let's consider the perspective of supply chains and economics. The global hardware industry depends on a sprawling web of raw material extraction, production hubs, and distribution networks, all of which are becoming increasingly vulnerable to disruptions.

Extreme weather events are predicted to rise in frequency and severity over the next two decades. Similarly, disruptions like the Suez Canal blockage of 2021 highlight the fragility of global shipping routes. As oceans grow rougher and waterways become less predictable, the economic calculus of long-haul transportation will shift dramatically.

This fragility points to the necessity of super-local decentralised production [15]. Reshoring—the process of bringing production closer to consumer markets—becomes not just an environmental imperative but a business necessity [16]. However, labor costs in developed nations are prohibitively high, and the workforce lacks the training for mass production at scale. The only viable path forward is to automate production to such an extent that it minimizes both labor input and logistics.

Closing the Loop on Resources

Even after hardware is produced, its lifecycle presents challenges. Devices at the end of their utility are often discarded, with precious metals and other materials poorly recovered. Current recycling methods struggle with low recovery rates for certain materials or render recovery more expensive than raw extraction. A closed loop, where resources are reclaimed and reused, will require a design philosophy that builds recyclability into the product from the outset. A potential answer could be in bacteria based bioreactors that could remove metals [17] from disposed devices and recycle it for production as efficiently as demonstrated by UC Berkeley Researcher Cecilia Martinez-Gomez.

Limitations and Challenges

The heart of modern electronics—silicon-based semiconductors—remains irreplaceable in the near term. The processes explored in this thesis are not equipped to fabricate silicon wafers but can build around them. However, these methods could extend the life of these active components by enabling easy extraction and reintegration of functional chips from obsolete devices to new ones. Further, volumetric additive manufacturing is still in its infancy. Speed, material diversity, and scalability remain barriers to adoption.

Governments must incentivize local production and recovery through subsidies, tax breaks, and stricter regulations on waste. Significant funding, interdisciplinary research, and global collaboration are essential.

What If?

The following speculative questions were formulated based on discussions with experts across various domains. These conversations opened up imaginative possibilities about the future of consumer electronics manufacturing.

What if you could walk into a store and order headphones that are custom-fitted to your ears and preferences, created on the spot by a volumetric production machine? What if these devices not only fit and sound better but also reduce waste through precise, tailored production?

What if augmentation devices were personalized to each individual's needs, eliminating the constraints of product generations and offering hardware updates as seamlessly as software ones?

What if, at the end of its useful life, your device could be returned, broken down in a bioreactor, and the materials reused — with the value given back to you as credit?

What if we developed polymer resins that aren't reliant on fossil fuels, possibly even sourced from local biomaterials?

What if the active electronics and conductors in our devices were made from graphene and carbon nanotubes, making the entire hardware stack printable and potentially turning used electronics into backyard compost or even carbon sinks?

What if Computer-Aided Engineering (CAE) design tools began to fully embrace the three-dimensional freedom offered by volumetric manufacturing, including the integration of generative design for both structure and circuits?

What if material utilization in product enclosures was optimized not for the lowest common denominator but tailored to each user's specific needs, thereby minimizing waste?

What if the future of hardware was as ubiquitous and disposable as paper, designed for quick design iterations and easy recycling? Rather than fighting the forces of convenience, we design for it.

What if we could print actuators and other robotic components, allowing machines to quite literally walk out of the manufacturing vat fully formed?



CONCLUSION

This thesis demonstrates the transformative potential of volumetric additive manufacturing in revolutionizing hardware production. Through experimentation at Berkeley's Design for Emerging and Nanoscale Manufacturing Lab, the research has validated the technical feasibility of volumetric printing of embedded electronics. While highlighting current technological limitations, particularly in semiconductor integration, this work reveals no fundamental barriers to realizing fully additive processes for hardware production.

The work challenges the prevailing narrative that sustainability can simply be retrofitted into existing design and production methods. The hardware industry's current approach fails to address the magnitude of the climate crisis. As global supply chains grow increasingly vulnerable to climate-related disruptions, from port closures to extreme weather events, the imperative for change extends beyond environmental concerns to basic business resilience. The proposed framework of volumetric manufacturing, combined with automated production and circular material recovery systems, offers a viable path forward that aligns environmental stewardship with hardware innovation.

The artifact, a functional concept for a volumetrically printed computer mouse serves as both proof-of-concept and a blueprint for future vision. By demonstrating the ability to create personalized, functional devices through resource-optimizing processes, the research challenges core assumptions about the necessity of mass production and global supply chains. The vision of super-local production, enabled by advanced manufacturing techniques, suggests a future where hardware can be produced on-demand, precisely tailored to user needs, and efficiently reclaimed at end-of-life—all while dramatically reducing environmental impact.

Looking forward, this thesis opens several critical areas for future research and development. Priority should be given to three key domains: advancing fossil-fuel-free polymer resins, developing generative design tools that fully exploit the three-dimensional freedom of volumetric manufacturing, advancing metal printing processes, and creating efficient material recovery processes through advanced chemical and biological methods. Success in these areas will require substantial investment, cross-disciplinary collaboration between materials scientists, engineers, and designers, and supportive policy frameworks that incentivize sustainable production methods. While significant challenges remain, this thesis establishes a foundation for meaningful conversation for change in the hardware industry. By bridging the gap between speculative design and technical feasibility, it invites researchers, industry leaders, and policymakers to collaborate in building a more sustainable and resilient future.



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